UNIT - III

FLEXURAL & SHEAR STRESSES

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- FLEXURAL STRESSES: theory of simple bending, derivation of the bending equation
- Bending stresses, section modulus of rectangular, circular, I and T sections
- Design of simple beam sections
- SHEAR STRESSES: derivation of the shear stress formula
- Shear stress distribution across rectangular, circular, triangular, I and T sections

Tip: $M/I = \sigma/y = E/R$ and $\tau = V A \bar{y} / (I b)$ are the two master equations of this unit.

1. Theory of simple bending

When a beam carries transverse loads, the bending moment makes it curve. The top fibres shorten (compression) and the bottom fibres lengthen (tension) for a sagging moment. Between them lies a layer that neither shortens nor lengthens - the *neutral surface*. Its trace on the cross-section is the *neutral axis* (N.A.).

Pure bending: a beam portion subjected to constant B.M. and zero shear force (e.g. the middle third of a beam loaded by two equal loads symmetric about the centre).

2. Assumptions in the theory of simple bending

- The material is homogeneous and isotropic.
- Hooke's law holds - stresses are within the elastic limit.
- E is the same in tension and in compression.
- Transverse sections plane before bending remain plane after bending.
- The beam is initially straight and has a constant cross-section, the radius of curvature is large compared with the depth.
- Each layer is free to expand or contract independently of the layers above and below.
- The beam is bent in a plane of symmetry, loads act in that plane.

3. Derivation of the bending equation

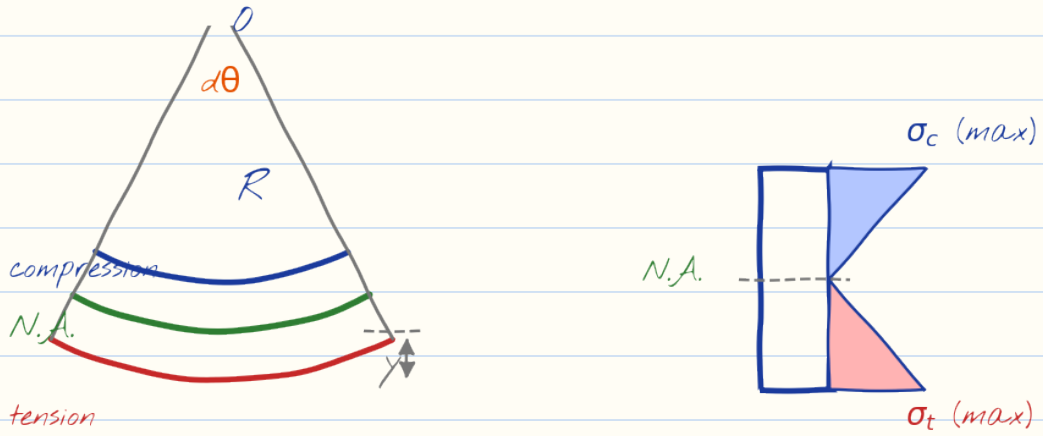


Fig: bending of a beam element, stress distribution on the section

Consider a small length of beam bent to a circular arc of radius R (to the neutral axis), subtending angle dθ at the centre O.

Length of N.A. layer = R dθ (unchanged)

Length of a layer at distance y below N.A. = (R + y) dθ

Strain $\varepsilon = [(R + y)\,d\theta - R\,d\theta] / R\,d\theta = y / R$

Stress $\sigma = E\,\varepsilon = E\,y / R$ i.e. $\sigma / y = E / R$ ...(1)

Stress varies linearly with y: zero at N.A., maximum at the extreme fibre.

Position of N.A.: no net axial force on the section, so

$$\int \sigma \, dA = (E/R) \int y \, dA = 0 \quad \Rightarrow \quad \int y \, dA = 0$$

i.e. the neutral axis passes through the centroid of the section.

Moment of resistance: moment of the force $\sigma \, dA$ about N.A. is $\sigma y \, dA$.

$$M = \int \sigma y \, dA = (E/R) \int y^2 \, dA = E \, I / R \qquad \ldots(2)$$

$$M / I = \sigma / y = E / R$$

M = bending moment (N·mm), I = M.I. of the section about N.A. (mm^4), σ = bending stress (N/mm^2), y = distance from N.A. (mm), R = radius of curvature (mm).

4. Section modulus

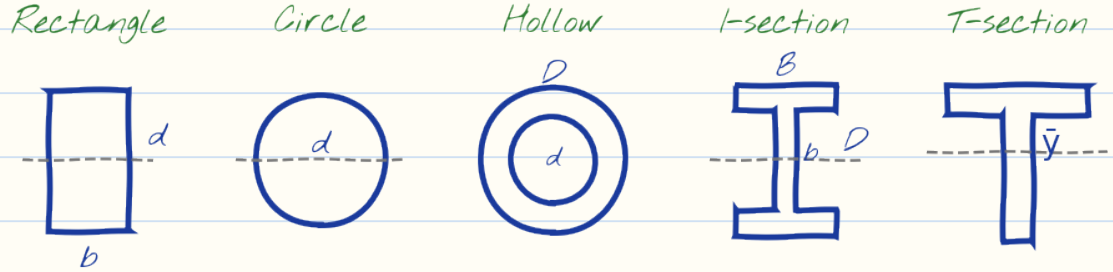
The maximum stress occurs at the extreme fibre, $y = y_{max}$:

$$\sigma_{max} = M \, y_{max} / I = M / Z \qquad Z = I / y_{max}$$

(section modulus, mm^3)

- Z measures the strength of a section in bending: $M_r = \sigma_{allow} Z$.
- For an unsymmetrical section (T, L) there are two moduli: $Z_{top} = I/y_t$ and $Z_{bottom} = I/y_b$.
- A section with more material far from N.A. (I-section) has a larger Z for the same area.

5. M.I. and section modulus of standard sections



Section	I about N.A.	$Z = I / y_{max}$
Rectangle $b \times d$	$b\,d^3 / 12$	$b\,d^2 / 6$
Solid circle, dia d	$\pi\,d^4 / 64$	$\pi\,d^3 / 32$
Hollow circle D, d	$\pi\,(D^4 - d^4) / 64$	$\pi\,(D^4 - d^4) / 32D$
Triangle b, h (about centroid)	$b\,h^3 / 36$	$b\,h^2 / 24$ (at base)
Symmetrical I-section	$(B\,D^3 - b\,d^3) / 12$	$I / (D/2)$

For I-section B = flange width, D = overall depth, b = (B – web thickness) and d = clear depth between flanges if you use $I = (B\,D^3 - b\,d^3)/12$ (the hollow-out method). For built-up sections use the parallel axis theorem $I = I_G + A\,h^2$.

6. Worked examples

Example 1: rectangular beam

A simply supported timber beam 100 mm × 200 mm, span 4 m, carries a u.d.l. w. Permissible bending stress = 8 MPa. Find w.

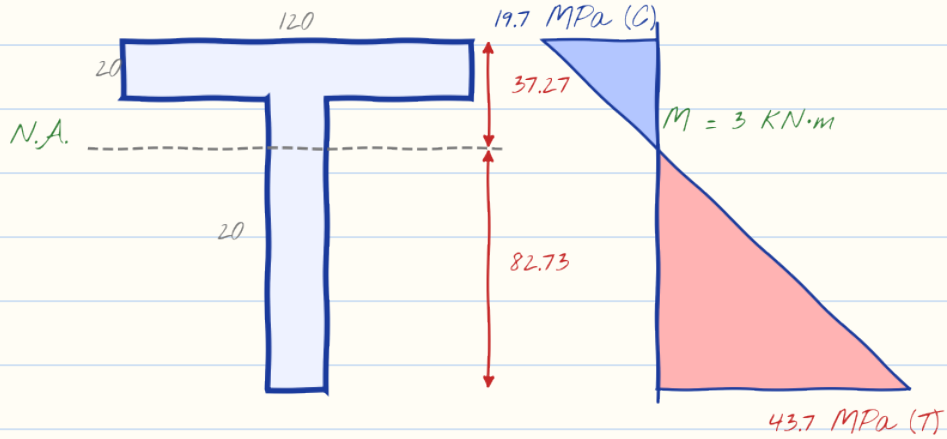
$Z = b\,d^2 / 6 = 100 \times 200^2 / 6 = 6.667 \times 10^5\ mm^3$

$M_{max} = \sigma\,Z = 8 \times 6.667 \times 10^5 = 5.333 \times 10^6\ N\cdot mm = 5.333\ KN\cdot m$

$w\,L^2/8 = 5.333 \Rightarrow w = 8 \times 5.333 / 4^2 = 2.67\ KN/m$

Example 2: T-section

T-section: flange 120 mm × 20 mm (top), web 20 mm × 100 mm. Find the bending stresses for a sagging moment of 3 KN·m.



$A_1 = 2400$ ($y = 10$ from top), $A_2 = 2000$ ($y = 70$)

$\bar{y} = (2400 \times 10 + 2000 \times 70) / 4400 = 37.27$ mm from top

$I = [120 \times 20^3/12 + 2400 \times 27.27^2] + [20 \times 100^3/12 + 2000 \times 32.73^2]$

$= 80\,000 + 1\,784\,880 + 1\,666\,667 + 2\,142\,400 = 5.674 \times 10^6$ mm^4

$\sigma_{top} = 3 \times 10^6 \times 37.27 / 5.674 \times 10^6 = 19.7$ MPa (compressive)

$\sigma_{bottom} = 3 \times 10^6 \times 82.73 / 5.674 \times 10^6 = 43.7$ MPa (tensile)

7. Design of simple beam sections

Design steps:

- 1. Find M_{max} from the B.M.D. (and include self-weight if required).
- 2. Required section modulus $Z_{req} = M_{max} / \sigma_{allow}$.
- 3. Choose the section so that $Z_{provided} \geq Z_{req}$.
- 4. Check shear stress, deflection and local buckling.

Rectangular ($d = 2b$): $b = (3Z / 2)^{1/3}$ Circular: $d = (32Z / \pi)^{1/3}$

Example 3: design

Design a rectangular ($d = 2b$) and a circular section for $M_{max} = 30$ KN·m and $\sigma_{allow} = 150$ MPa.

$Z_{req} = 30 \times 10^6 / 150 = 2 \times 10^5$ mm^3

Rectangular: $b = (1.5 \times 2 \times 10^5)^{1/3} = 66.9$ mm $\Rightarrow$ 70 mm x 140 mm ($A = 9800$ mm^2)

Circular: $d = (32 \times 2 \times 10^5 / \pi)^{1/3} = 126.8$ mm $\Rightarrow$ $d = 130$ mm ($A = 13\,273$ mm^2)

For the same strength the circular section needs about 35 % more material. An I-section is more economical still - most material is placed far from the N.A.

8. Shear stresses in beams – derivation

A transverse shear force V causes shear stress τ on the cross-section. By complementary shear the same τ acts on horizontal planes. Consider a layer at distance y from the N.A.

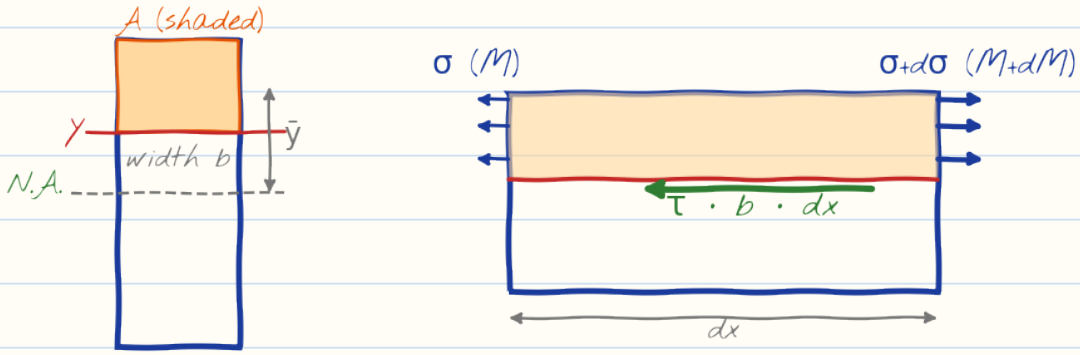


Fig: element of length dx, shaded area A above the level y

Force on the left face of the shaded part = $\int \sigma \, dA = (M/I) \int y \, dA$

Force on the right face = $((M + dM)/I) \int y \, dA$

Difference = $(dM/I) \int y \, dA = (dM/I) A \bar{y}$ (resisted by shear on the bottom face)

$\tau \, b \, dx = (dM/I) A \bar{y}$ and $dM/dx = V$

$$\tau = V A \bar{y} / (I b)$$

- V = S.F. at the section, A = area of the section <u>above</u> (or below) the level considered,
- $\bar{y}$ = distance of the centroid of A from the N.A., I = M.I. of whole section about N.A.,
- b = width of the section at the level considered.
- τ is maximum at the N.A. in most sections and zero at the extreme fibres.

9. Shear stress distribution

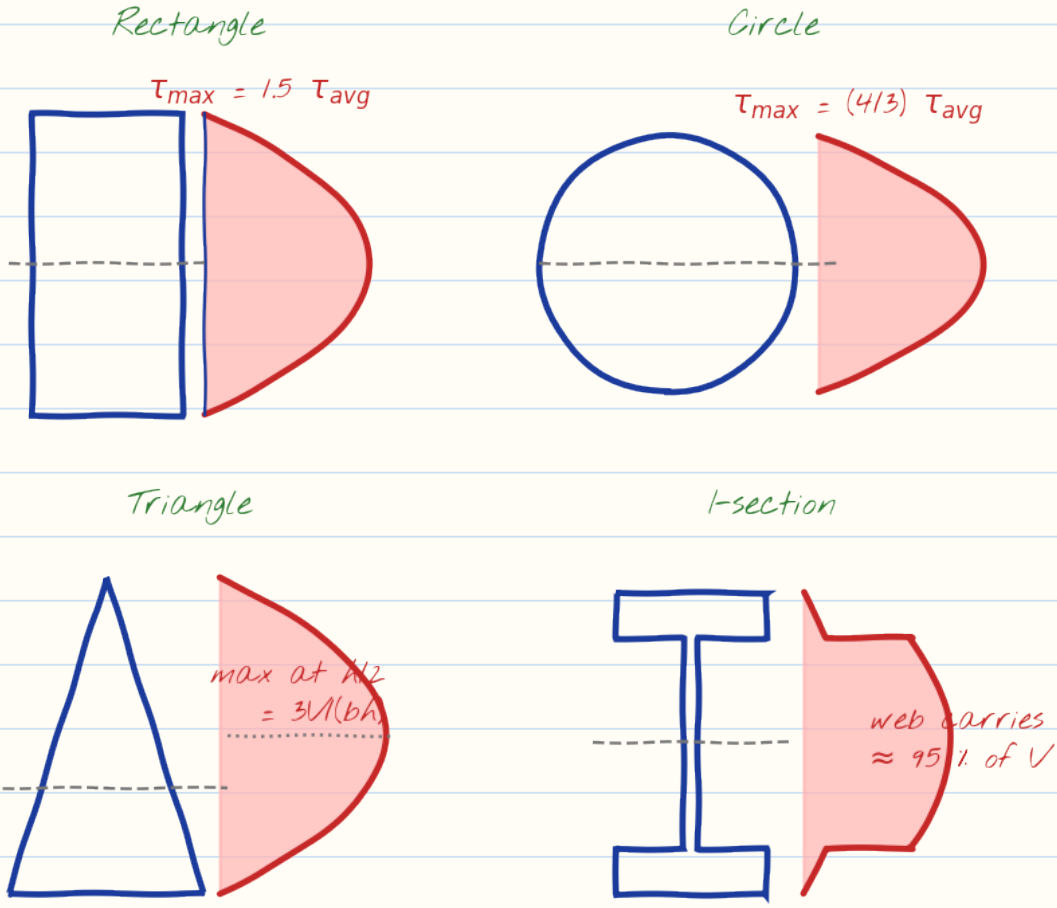


Fig: Shear stress distributions

(a) Rectangular section b x d

$\tau = (V/2I)(d^2/4 - y^2)$ - parabolic.

$\tau_{max} = 3V / 2A = 1.5\ \tau_{avg}$ (at N.A.) $\tau = 0$ at $y = \pm d/2$

(b) Circular section, diameter d

$\tau = (4V / 3\pi R^2)(1 - y^2/R^2)$ - parabolic.

$\tau_{max} = 4V / 3A = 1.333\ \tau_{avg}$ (at N.A.)

(c) Triangular section (base b, height h, apex up)

$\tau = 12\ V y (h - y) / (b\ h^3)$ (y measured from apex).

$\tau_{max} = 3V / (b\ h) = 1.5\ \tau_{avg}$ at h/2 from apex; at N.A. (2h/3): $\tau = 8V / 3bh = 1.33\ \tau_{avg}$

The maximum shear stress in a triangle is NOT at the neutral axis - it occurs at mid-height.

(d) I-section (flange B, depth D, web b, clear web depth d)

- In the web the variation is parabolic, in the flanges it is small (flange is wide).
- At the web-flange junction there is a sudden jump (the width changes from B to b).
- τ at N.A. (max) = $V / (8 / b) [B (D^2 - d^2) + b d^2]$
- τ at junction (in web) = $V B (D^2 - d^2) / (8 / b)$, inside the flange = (b/B) of that value.
- The web carries about 95 % of the shear force; flanges carry most of the bending moment.

(e) T-section

Find N.A. ($\bar{y}$), then use $\tau = V A \bar{y} / (I b)$ at the level required. Maximum τ at the N.A.; for the flange-web junction compute two values (with b = flange width, and b = web thickness) - a jump occurs.

10. Worked examples on shear

Example 4: rectangular beam

A rectangular beam 100 mm x 200 mm carries a shear force of 30 KN. Find τ_{avg} and τ_{max}.

$\tau_{avg} = 30000 / (100 \times 200) = 1.5$ MPa

$\tau_{max} = 1.5 \times 1.5 = 2.25$ MPa at the neutral axis

Example 5: I-section

I-section: flanges 100 mm x 20 mm, web 10 mm thick, overall depth 200 mm. V = 100 KN. Find shear stresses at N.A. and at the web-flange junction.

$I = (100 \times 200^3 - 90 \times 160^3)/12 = (8 \times 10^8 - 3.686 \times 10^8)/12 = 3.595 \times 10^7$ mm^4

At N.A.: $A \bar{y} = 100 \times 20 \times 90 + 10 \times 80 \times 40 = 212\,000$ mm^3

$\tau_{NA} = 100000 \times 212000 / (3.595 \times 10^7 \times 10) = 59.0$ MPa

At junction (web side): $A\bar{y} = 180\,000 \Rightarrow \tau = 100000 \times 180000 / (3.595 \times 10^8) = 50.1$ MPa

Just inside the flange: $\tau = 50.1 \times 10/100 = 5.0$ MPa (large jump!)

Average web shear = 100000 / (10 × 200) = 50 MPa – close to $\tau_{max} = 59$ MPa.

More solved problems

Example 6: bending stress in a circular beam

A simply supported beam, span 3 m, circular section d = 100 mm, central point load 10 KN. Find the maximum bending stress.

$M_{max} = W L / 4 = 10 \times 3 / 4 = 7.5$ KN·m $= 7.5 \times 10^6$ N·mm

$Z = \pi d^3 / 32 = \pi \times 100^3 / 32 = 98\,175$ mm^3

$\sigma_{max} = M / Z = 7.5 \times 10^6 / 98\,175 = 76.4$ MPa

Example 7: radius of curvature

A steel wire of 5 mm dia is bent round a drum of radius 2.5 m. $E = 2 \times 10^5$ MPa. Find the maximum stress in the wire.

$\sigma / y = E / R \Rightarrow \sigma = E y / R = 2 \times 10^5 \times 2.5 / 2500 = 200$ MPa

Here the radius is to the N.A. (use R + d/2 if the drum radius is to the inner surface).

Example 8: hollow circular section

Hollow shaft-type beam D = 100 mm, d = 80 mm carries M = 5 KN·m. Find σ_{max}.

$I = \pi (100^4 - 80^4) / 64 = \pi \times 5.904 \times 10^7 / 64 = 2.898 \times 10^6$ mm^4

$Z = I / 50 = 57\,962$ mm^3

$\sigma_{max} = 5 \times 10^6 / 57\,962 = 86.3$ MPa

Example 9: bending + shear check

S.S. beam, span 6 m, u.d.l. 20 KN/m, rectangular section 150 mm × 300 mm. Find σ_{max} and τ_{max}.

$M_{max} = w L^2 / 8 = 20 \times 36 / 8 = 90$ KN·m; $V_{max} = w L / 2 = 60$ KN

$Z = 150 \times 300^2 / 6 = 2.25 \times 10^6$ mm^3 $\Rightarrow$ $\sigma_{max} = 90 \times 10^6 / 2.25 \times 10^6 = 40$ MPa

$\tau_{max} = 1.5 \times 60000 / (150 \times 300) = 2.0$ MPa

Bending stress governs in slender beams; shear matters for short, deep beams and thin webs.

Example 10: circular section shear

A circular beam of dia 100 mm carries $V = 50$ KN. Find τ_{avg} and τ_{max}.

$A = \pi/4 \times 100^2 = 7854$ mm^2, $\tau_{avg} = 50000 / 7854 = 6.37$ MPa

$\tau_{max} = (4/3) \times 6.37 = 8.49$ MPa at the neutral axis

Comparison of sections

Section (same area)	Relative Z	Remark
Circle	1.00	least efficient
Square (sides vertical)	≈ 1.18	better than circle
Rectangle d = 2b	≈ 1.67	deep, narrow beams
I-section	3 to 5	most economical

- Flexural rigidity = E I. A deeper section increases I with the cube of depth.
- Rectangular beam: placing it with depth vertical (d > b) gives a much stronger beam than flat.
- Section modulus Z, not I alone, decides the moment a section can carry.

Example 11: shear in the T-section of Example 2

Same T-section ($I = 5.674 \times 10^6$ mm^4, N.A. 37.27 mm from top), $V = 20$ KN. Find τ at N.A. and at the flange-web junction.

Below N.A. (easier): web depth 82.73 mm, $A = 20 \times 82.73 = 1654.6$ mm^2, $\bar{y} = 41.37$ mm

$A\bar{y} = 68\,440\ mm^3 \Rightarrow \tau_{NA} = 20000 \times 68440 / (5.674\times10^6 \times 20) = 12.06\ MPa$

Junction: A = flange $2400\ mm^2$, $\bar{y} = 27.27\ mm$, $A\bar{y} = 65\,448\ mm^3$

in the web ($b = 20$): $\tau = 20000\times65448 / (5.674\times10^6\times20) = 11.53\ MPa$

in the flange ($b = 120$): $\tau = 11.53 \times 20/120 = 1.92\ MPa$

Practice problems (answers given)

- Rectangular beam 80 x 160 mm carries $M = 6\ KN\cdot m$. Find σ_{max}. Ans: $Z = 3.413\times10^5\ mm^3$, $\sigma = 17.6\ MPa$
- Solid circular beam $d = 60$ mm, permissible stress 100 MPa. Find the moment of resistance. Ans: 2.12 KN·m
- Rectangular beam 60 x 120 mm, $V = 18\ KN$. Find τ_{avg} and τ_{max}. Ans: 2.5 MPa, 3.75 MPa
- If $\sigma_{max} = 120\ MPa$ occurs at $y = 100$ mm, find σ at $y = 40$ mm. Ans: 48 MPa
- A steel strip 2 mm thick is bent to a radius of 1 m. Find the maximum stress ($E = 200\ GPa$). Ans: $\sigma = Ey/R = 2\times10^5 \times 1/1000 = 200\ MPa$

Important questions

- State the assumptions of simple bending and derive $M/I = \sigma/y = E/R$.
- Show that the neutral axis passes through the centroid of the section.
- Define section modulus, obtain it for rectangular, circular and hollow circular sections.
- Derive the shear stress formula and draw the distribution for a rectangular section.

- Draw the shear stress distribution for an I-section and explain the jump at the junction.
- A T-beam is loaded as shown - find the maximum tensile and compressive bending stresses.

Quick revision - Unit III

$$M/I = \sigma/y = E/R \qquad Z = I/y_{max} \qquad M = \sigma Z \qquad \tau = \frac{V A \bar{y}}{I b}$$

Section	τ_{max} / τ_{avg}	Location
Rectangle	1.5	N.A.
Circle	4/3	N.A.
Triangle	1.5	h/2 from apex
I-section	≈ 1.1 - 1.2	N.A.

- Important: derive $M/I = \sigma/y = E/R$ with assumptions.
- Important: derive $\tau = V A \bar{y} / (I b)$ and plot shear distribution for rectangular, I and T sections.
- Important: design problems; bending and shear stress problems on I and T sections.